

FCU3

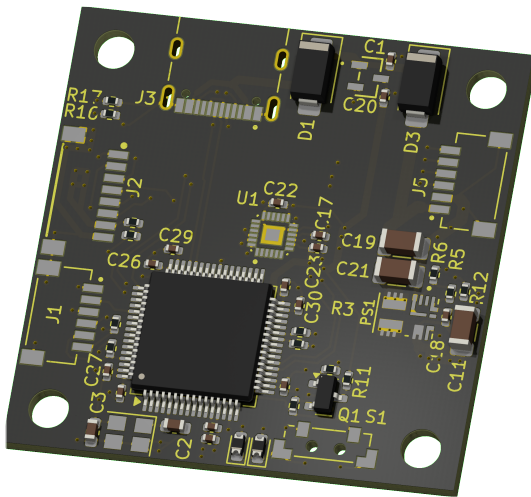
Variant: CHECKED

2025-12-10

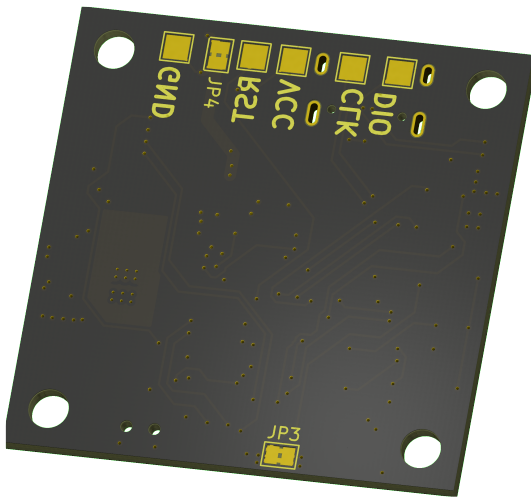
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 10-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

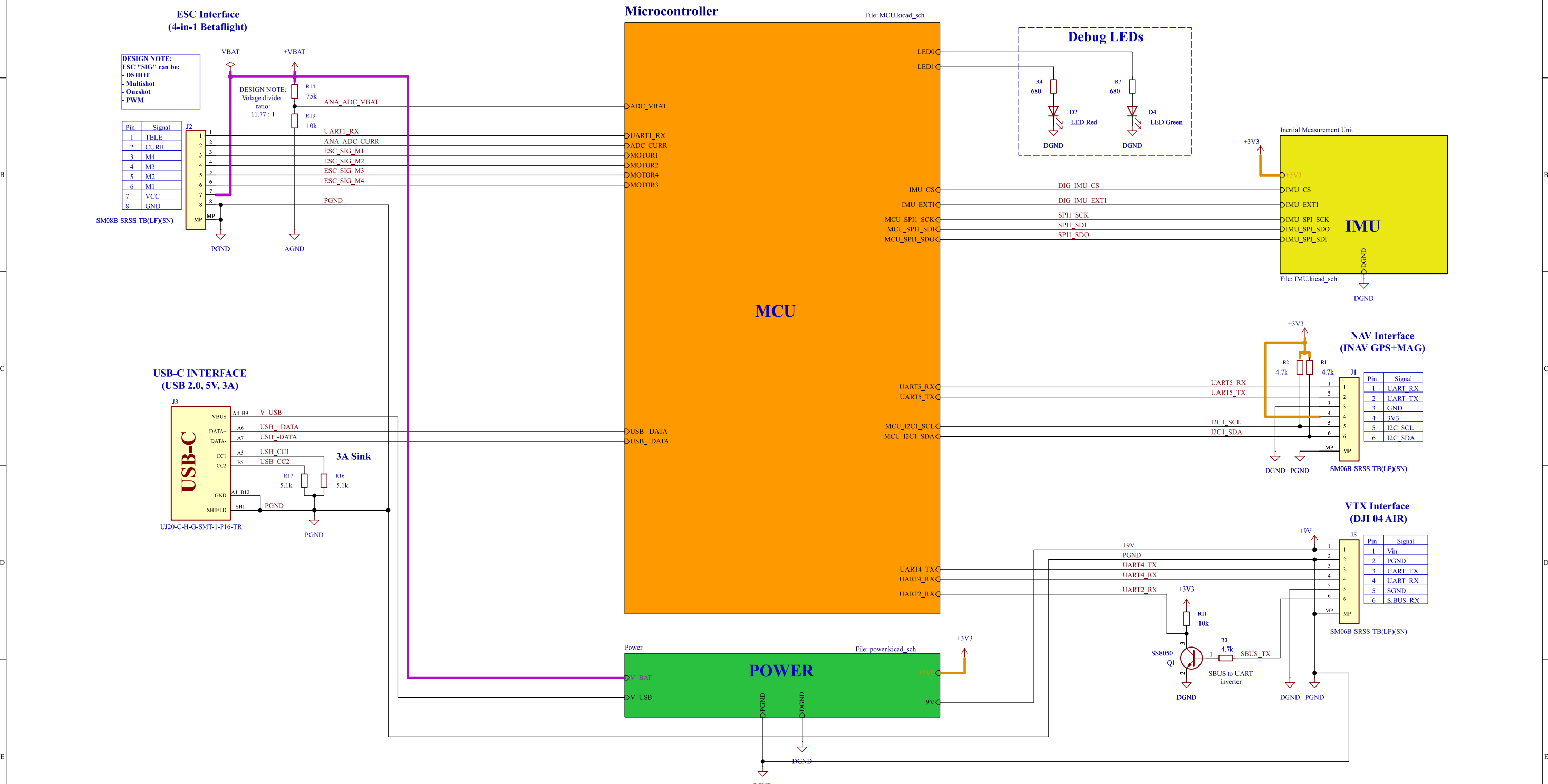
LAYOUT NOTE:

Example text for critical layout guidelines.

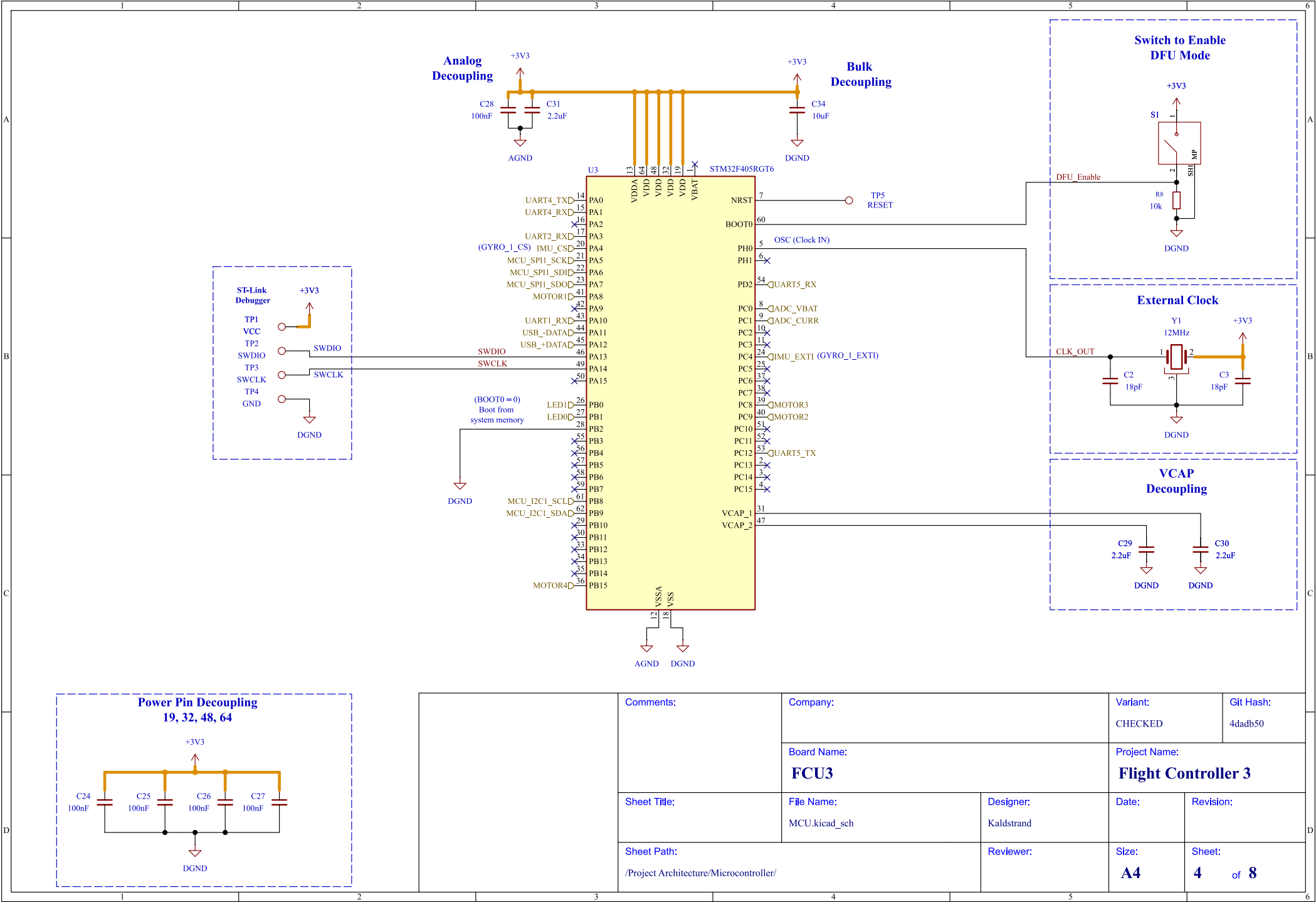
	Comments:	Company:	Variant:	Git Hash:
		Kaldstrand Electronics	CHECKED	4dadb50
	Sheet Title:	Board Name:	Project Name:	
		FCU3	Flight Controller 3	
	File Name:	Designer:	Date:	Revision:
	FCU3.kicad_sch	Kaldstrand	2025-01-12	+(Unreleased)
Sheet Path:	Reviewer:	Size:	Sheet:	
/		A3	1 of 8	

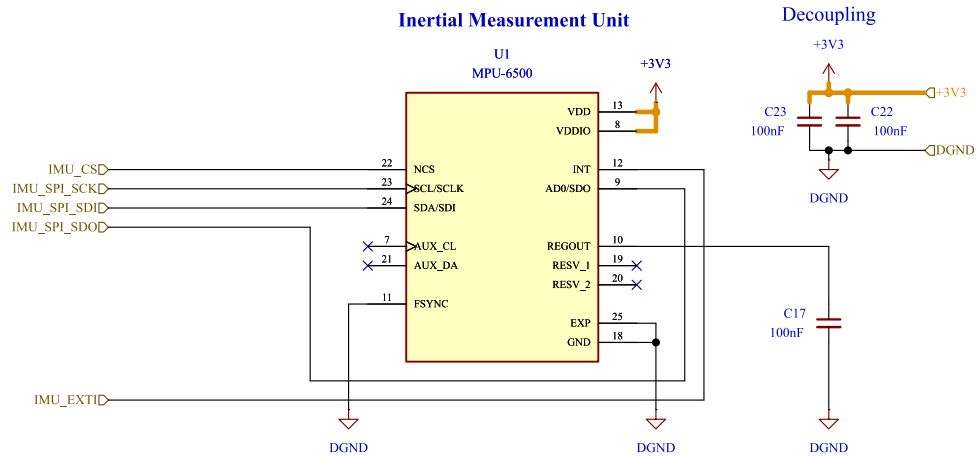
[illegible]

[3] Project Architecture



Grounding Strategy Ground Ties DGND JP3 AGND DGND AGND Mounting Holes H1 H2 H3 H4		PGND				Comments:		Company: Kaldstrand Electronics		Variant: CHECKED		Git Hash: 4dadb50	
						Board Name: FCU3				Project Name: Flight Controller 3			
Sheet Title: Project Architecture		File Name: Project Architecture.kicad_sch		Designer: Kaldstrand		Date: 2025-01-12		Revision: + (Unreleased)					
Sheet Path: /Project Architecture/		Reviewer:		Size: A3		Sheet: 3 of 8							





	Comments:	Company:	Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4 Sheet: 5 of 8

To-Do:

- Add all settings used in Vbat to 9V converter.
- Add all settings used in 3.3V LDO.

```

    graph LR
      VB[V_BATD] --> U1[10-26V to 9V2A Step-Down Converter  
File: stepdown_10-26vin_9v2a.kicad_sch]
      U1 -- "+9V" --> P9V[+9V]
      U1 --- GND1[GND]
      US[V_USBD] -- "5V @ (0, 500, 3000)mA" --> D1[SS14-M3/SAT  
D1]
      US -- "5V @ (0, 500, 3000)mA" --> D3[SS14-M3/SAT  
D3]
      D1 -- "10-25.2V @ (0, 3, 100)A" --> LDO_VIN[VIN]
      D3 -- "10-25.2V @ (0, 3, 100)A" --> LDO_VIN
      LDO_VIN --> LDO[U2  
AP7375-33SA-7  
LDO  
3V3 @ (0, 50, 200)mA]
      LDO --- GND2[GND]
      LDO -- "VOUT" --> P3V3[+3V3]
      P3V3 --> C20[C20  
1uF]
      C20 --- DGND[DGND]
      PGND[PGND] --- JP4[JP4]
      DGND --- JP4
  
```

DESIGN NOTE:
JP4 is the Power Ground (PGND) to Digital Ground (DGND) connection.

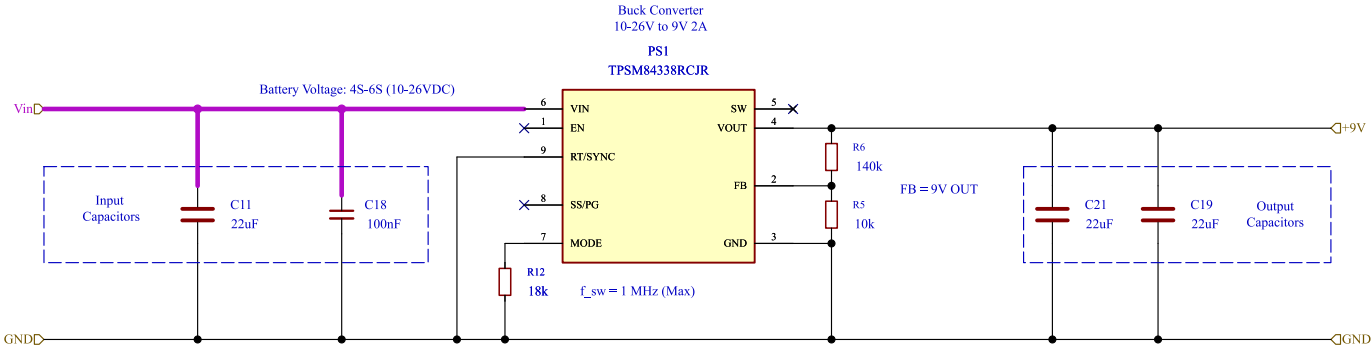
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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: power.kicad_sch	Designer: Kaldstrand	Date:	Revision:
	Sheet Path: /Project Architecture/Power/		Reviewer:	Size: A4	Sheet: 6 of 8

D				Comments:	Company:		Variant:	Git Hash:
						CHECKED	4dad50	
					Board Name: FCU3		Project Name: Flight Controller 3	
				Sheet Title:	File Name: power.kicad_sch	Designer: Kaldstrand	Date:	Revision:
				Sheet Path: /Project Architecture/Power/		Reviewer:	Size: A4	Sheet: 6 of 8
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DESIGN NOTE:
 $R_{FBB} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBB} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBB} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:		Variant:	Git Hash:
				CHECKED	4dad50
		Board Name:		Project Name:	
		FCU3		Flight Controller 3	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
		stepdown_10-26vin_9v2a.kicad_sch	Kaldstrand		
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Power/10-26V to 9V2A Step-Down Converter/		A4	8 of 8	

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8